Drier soils delay plant phenology across climate change experiments in temperate forest and grassland systems

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Abstract

Previous meta-analyses of phenology responses to climate change have focused largely on temperature as a driver of observed shifts. Yet climate change also affects soil moisture, which is limiting to many biological responses. Here we synthesize microclimate and phenology data from climate change experiments to quantify how interactions between soil moisture and temperature affect plant phenology in temperate forests and grasslands. We find that plants delay phenology (budburst, leafout and flowering) in drier soils, with the largest delays seen in budburst. Standardized effect sizes of soil moisture were much smaller than for temperature (-1.7 days versus -7.9 days, respectively) on average, but with variability across species in the response to soil moisture. This species-level variability combined with forecasts of increased drought in some locations suggests soil moisture will become increasingly important to future phenology in certain systems. Supporting this, we show that forecasted soil moisture changes could have important effects on the phenology of some species, with drying potentially muting advances due to warming alone. Our results highlight that soil moisture plays an important role in the phenology of temperate systems, and thus is likely to alter ecosystem functions tied to phenology, such as growth and carbon storage. Incorporating local context, including ecological knowledge of relevant species and down-scaled climate change projections, will be critical for planning appropriate management and conservation in response to phenology shifts and related ecosystem services.

INTRODUCTION

Climate change is affecting organisms by altering conditions such as temperature and soil moisture around the world (Parmesan, 2006; Chen et al., 2011). Some of the most widespread biological responses to climate change are altered phenology, the timing of recurring biological events, which have shifted 2.3-5.1 days per decade (Parmesan, 2006; Poloczanska et al., 2013; Root et al., 2003). Shifts in plants are the most widely documented, with spring phenology (budburst, leafout, and flowering) occurring earlier in recent years (Wolkovich et al., 2013), and senescence occurring later (Taylor et al., 2008; Delpierre et al., 2009).

Phenology shifts are typically attributed to warming temperature, a known and well-studied driver of plant phenology (e.g., Cleland et al., 2007; Piao et al., 2019). The timing of spring budburst, for example, depends on temperature, advancing with both increased chilling (the prolonged exposure to cold temperatures after growth cessation in the fall) and forcing (exposure to warm temperatures). Forcing effects are typically considered more dominant, so much so that many models use only forcing to predict phenology. These include common models of 'growing degree days' (GDD), in which phenology events are triggered after a certain thermal sum is reached (e.g., Olsson and Jönsson, 2014). Recent trends of advancing spring phenology have generally attributed this advance to warming springs—and potentially winters—with myriad studies focused on how temperature determines spring phenology (Fujisawa and Kobayashi, 2010; Ibanez et al., 2010; Cook et al., 2012).

Effects of changing patterns of precipitation and soil moisture on plant phenology have received less attention, but fundamental plant biology suggests they should play an important role. High water-stress can slow budburst by inhibiting cell elongation (Essiamah and Eschrich, 1986), suggesting woody plant spring phenology is fundamentally limited by water even if this is rarely observed. Similarly, in grasslands, growing season start can be delayed by drought Cui et al. (2017). For plants already growing, however, drought conditions can advance flowering (Hamann et al., 2018), as it is beneficial for plants to complete fruiting before water becomes too limiting.

Though many climate change experiments have focused on effects of altered precipitation regimes, the role of soil moisture on phenology has been most reported in arid and grassland ecosystems (e.g., Essiamah and Eschrich, 1986; Reich and Borchert, 1984; van Schaik et al., 1993; Tao et al., 2019). Effects in arid systems

are diverse, often with no overall shift or trend in phenology for most single experiments (e.g., Sherry et al., 2007; De Kauwe et al., 2017; Howell et al., 2020), as well as meta-analyses (Lu et al., 2023). Results in mesic regions have been equally diverse: Wang et al. (2022*b*) found that decreasing precipitation frequency correlates with earlier leafout in many regions, while others have found variation in moisture sensitivity across ecoregions (Seyednasrollah et al., 2018). Increasing research using large-scale observational phenology data (e.g., remote sensing products such as NDVI) has documented the importance of soil moisture from forests to grasslands (Lian et al., 2020; Shen et al., 2022; Liu et al., 2024), and suggest temperature may play a role through moderating soil moisture (Liu et al., 2024). Teasing out the role of soil moisture from temperature is challenging through long-term climate trends alone, however.

Field-based climate change experiments that warm plots to different levels and apply precipitation or drought treatments could help tease apart the effects of temperature and moisture on plant phenology. Experiments can combine temperature and precipitation treatments to decouple them compared to what may be observed in nature, allowing their effects to be more robustly quantified. Additionally, experiments that monitor daily soil moisture and air temperature at the plot-level can help translate how these treatments alter microclimates to affect plant phenology (Ettinger et al., 2019). While previous meta-analyses of phenology in climate change experiments have focused primarily on effects of temperature (e.g., Wolkovich et al., 2012), or compared treatment effects (Zhou et al., 2023), there has been little synthetic work on moisture effects across experiments.

Here we use measured microclimate and phenology data from published experiments (Fig. S1) to test how soil moisture and above-ground temperature together affect plant phenology (budburst, leafout, flowering). Our aims were to: (1) quantify effects of soil moisture versus temperature alone and synergistically across species; (2) evaluate how consistent effects were across species, functional groups and biomes (forest versus grassland), and (3) forecast effects to understand future implications of moisture shifts with warming for phenology.

MATERIALS AND METHODS

Data– To investigate how soil moisture interacts with temperature to affect phenology, we used two databases that compiled data from climate change experiments. Microclimate data came from the MicroClimate from Climate Change Experiments (MC3E) database (Ettinger and Wolkovich, 2018; Ettinger et al., 2019). Phenology data came from a ExPhen, a database of phenology from climate change experiments (which has not been previously analyzed Ettinger and Wolkovich, 2022).

Both databases were created by first identifying published, active-warming field experiments, many of which included precipitation manipulations. We focused on *in situ* active-warming manipulations because recent analyses indicate that active-warming methods are the most controlled and consistent methods available for experimental warming (Kimball, 2005; Kimball et al., 2008; Aronson and McNulty, 2009; Wolkovich et al., 2012). We carried out a full literature review to identify potential active-warming field experiments, following the methods and search terms of Wolkovich et al. (2012) for their Synthesis of Timings Observed in iNcrease Experiments (STONE) database (Wolkovich et al., 2013), but restricting our focus to active-warming experiments. Further, because our goal was to tease out variation in microclimate (including temperature and soil moisture), we focused on warming studies that included multiple levels of warming and/or precipitation treatments. These additional restrictions constrained the list to 11 new studies published after the STONE database, as well as six of the 37 studies in the STONE database. We contacted authors to obtain daily microclimate and phenology data for these 17 studies and received data (or obtained publicly available data) for 10 of them, as well as datasets from five additional sites offered or suggested to us over the course of our literature review and data analysis. The daily temperature and soil moisture data from these 15 experiments comprise the MC3E database (Ettinger and Wolkovich, 2018; Ettinger et al., 2019). Of these, we were able to obtain plot-level phenology data from 14 experiments, which comprise the ExPhen database of experimental phenology, available at KNB (Ettinger and Wolkovich, 2022).

Here, we analyze phenology data from the eight experiments in ExPhen that contain both regularly monitored plot-level soil moisture and above-ground temperature data (Table S1, Fig. S1). To help examine variation among species and across sites, we focus on the most common phenophases monitored, which were measured in three or more different experiments: budburst, leafout, and flowering. Two of the eight experiments were

located in grassland ecosystems; the remaining six were in forests (Table S1). The database is species-rich, including 41 species monitored for budburst across five sites, 137 for leafout (across five sites), and 124 for flowering (across all eight sites), for a total of 190 species. These species span grasses (16 species), forbs (109 species), shrubs (29 species), and trees (36 species).

Analysis– To understand how soil moisture interacts with temperature to affect phenology, we fit models with microclimate predictor variables of measured soil moisture, measured above-ground temperature, and their interaction to phenology response data (budburst, leafout, flowering day of year). We excluded conifers from the analysis, because their phenology has distinct differences from angiosperm phenology Polgar et al. (2014) and conifer data existed from only one site in the database. For all phenophases, the response variable was day of year of the phenology event.

Predictors for our primary models were: measured plot-level above-ground temperature, soil moisture, and their interaction. We chose to use measured microclimate as explanatory variables, rather than categorical treatment levels or target warming level, in our meta-analysis because experimental treatment effects from warming and drought can interact to alter microclimate conditions, in part due to feedbacks between temperature and soil moisture conditions (Ettinger et al., 2019; McDaniel et al., 2014).

We used hierarchical Bayesian models to test for effects for each species, as well as an overall effect, while accounting for site, year and plot-level effects. Grouping factors (often called 'random effects') for all phenology models were species (with random slopes and intercepts), site (random intercept), and year nested within site (random intercept). We fit models using the programming language `Stan` (Carpenter et al., 2017) (`www.mc-stan.org`), accessed via the `brms` (Bürkner, 2021) package in R (R Core Team, 2022), version 4.1.3. For each model fit, we ran four chains simultaneously, each with 4 000 iterations (2 000 of which were used for warm-up). Equations for these models can be found in the Supplemental Methods.

Given our aim to directly compare moisture and temperature effects, we used standardized predictors, which have an added benefit of improving model stability (Gelman and Hill, 2007). Standardizing predictors is a common technique in regression analysis; here we z-scored predictor variables (subtracting the mean and dividing by the standard deviation) and report coefficients from standardized predictor models as per SD (standard deviation), alongside estimates of coefficients in their natural units. We report 80% uncertainty intervals parenthetically throughout the manuscript, unless otherwise stated, and we report on and interpret

effects when they do not cross zero at this interval, though almost always this extends to higher intervals (e.g., 90%). We consider effects 'weak' when they do not cross zero at 50% uncertainty intervals (but do at 80%) and strong when they do not cross zero using 80% uncertainty intervals.

RESULTS

We found that both higher soil moisture and higher temperatures advance phenology, meaning two common effects of warming experiments—soil drying and warming—have contrasting effects on phenology (Fig. 1, Table S2). Soil drying generally delays phenology and warming temperatures generally advance phenology. For budburst, wetter soils and warmer temperatures alone advanced phenology by -1.7 (-2.5,-1) days per SD of soil moisture (or -5.21 days per 10 percent increase in volumetric water content) and -7.9 (-10.5,-5.4) days per SD of temperature (or -3.4 per degree Celsius). We found a weak positive (antagonistic, in this case) interactive effect of soil moisture and temperature on budburst phenology (0.5 [-0.1-1.1] days per SD).

The magnitude of soil moisture effects varied across phenophases, with effects on budburst (-1.7 [-2.5,-1] days per SD increase in soil moisture) being stronger than leafout (-0.4 [-1.7,0.7] per SD increase in soil moisture) but similar to flowering (-1.3 [-2.5,-0.1] per SD increase in moisture). Similar to budburst, temperature effects were stronger than soil moisture for leafout (for which the temperature effect was -10.4 [-12.2,-8.5] days per SD increase in temperature) and flowering (for which it was -8 [-9.7,-6.3] per SD increase in temperature), across all species (Fig. 1). Estimates of interactions between soil moisture and temperature on phenology also varied by phenophase, with weakly positive effects estimated for leafout (0.5) [-0.4,1.4] and budburst (0.5 [-0.1,1.1]) and a strong negative interaction for flowering (-1.1 [-2.1,-0.2]). This negative interaction implies that there may be synergistic effects of soil moisture and temperature (both of which also have negative estimated effects on flowering), resulting in flowering that advances even more strongly than would be expected by simply adding together the estimated effects of temperature and moisture each acting alone.

These overall effects varied widely across species (Fig. 1, Table S2). Species-level variance for the effect of moisture was 2.7 (2.1,3.5) days for budburst, 3.8 (2.6,5.3) for leafout, and 3.8 (2.5,5.2) for flowering. Species-level variance was even greater for temperature effects: 11.4 (9.5,13.5) for budburst, 10.2 (8.6,12) days for leafout, and 6.2 (4.8,7.7) for flowering. Analysis of posterior distributions suggests that species-level

variability in responses to moisture is not predictable by life form (trees, shrubs, herbs, grasses, Fig. 2, column 2) nor by ecosystem (grassland versus forests, Fig. S2), across the three phenophases we studied. We did, however, observe more negative effects of temperature on trees compared to shrubs for budburst, and on both trees and grasses compared to shrubs and forbs for leafout (Fig. 2B,E,H). Interactions between temperature and moisture effects on leafout also skewed more positive for grasses compared to other life-forms (Fig. 2C,F,I).

We applied the budburst model (Fig. 1A, Table S2) to understand possible effects of future warming on phenology, under varying soil moisture conditions. Based on the estimated effects, wetter soils advanced spring budburst at a rate of 5 days per 10% increase in soil volumetric water content (VWC). Thus, if soil moisture is reduced by 10% of its current state, as is expected over the next 50 years in areas where many of the experiments were conducted (the northeastern United States) (Berg et al., 2017) (moving from, e.g., 21.5% VWC – the mean value for January-March across all sites for which budburst was monitored – to 19.4% VWC), budburst would be delayed by approximately 1 day on average, due to changes in soil moisture alone, and based on overall estimates across species. Soil moisture is highly variable in space as well as time, however (Vergopolan et al., 2022).We wanted to understand how this wide range of soil moisture conditions may lead to differences in phenology responses to warming with climate change, so we forecasted shifts in budburst under a "warming only" scenario, as well as a "very dry" scenario (with soil moisture conditions reduced by 100% compared to the average conditions in experiments included in the budburst dataset) and a "very wet" scenario (with soil moisture conditions increased by 100% compared to current conditions). This forecasting reveals wide variation across species in response to warming and changes in soil moisture (Fig. 3A). For example, for *Quercus nigra* and other species for which the estimated soil moisture - temperature interaction is small, phenological shifts with warming are similar in magnitude under both wet and dry conditions (Fig. 3B). However, for *Carya glabra*, a species for which soil moisture has a large effect () and a large negative interaction with temperature (), the difference in budburst between the wet and dry scenario is large and this difference increases with warming (Fig. 3C). *Nyssa sylvatica* exemplifies a species for which the interaction of soil moisture and temperature is dampening rather than synergistic, (Fig. 3D).

DISCUSSION

Synthesizing results from climate change experiments we found that soil moisture has important impacts on plant phenology in temperate ecosystems. While effects of water availability on phenology have been often reported in other systems, these have been generally confined to arid systems, where the effect of moisture on plant development is quite clear (e.g., Reich and Borchert, 1984; van Schaik et al., 1993; Cleverly et al., 2016; Bertiller et al., 1991), and alpine systems, where high snowpack can limit the length of growing season (effectively limiting when temperature can trigger development). Our results extend this across temperate forest and mesic grassland systems where the role of soil moisture has generally been considered much less due to the expectation that it is rarely limiting.

By leveraging phenology responses to plot-level soil moisture across multiple experiments, we show that soil moisture plays a potentially overlooked role phenology in temperature systems, especially for budburst and flowering, and could become increasingly important with continued climate change. In particular, our findings that dry soils delay phenology and that responses vary widely among species have major implications for forecasting and understanding existing experimental and observational results. These results support growing evidence from small-scale precipitation manipulations in forests (e.g., Currier and Sala, 2022) and observational and experimental studies in temperate grasslands (Liu et al., 2022; Seyednasrollah et al., 2018). Taken together our results show that moisture can be a hidden, but potentially limiting, factor affecting phenology in temperate systems not typically thought to be water-stressed ((Ettinger et al., 2019)).

High variation in responses to soil moisture across species and phenophases

Despite the overall delaying effect of soil drying that we found for phenology, our results suggest that forecasts will need to contend with high variation in species responses, as well as differences across phenophases 1. This might suggest that predicting variable responses to moisture will be challenging, especially since our results indicate there may not be readily apparent differences in soil moisture responses by broad functional types. This finding contrasts with some previous studies, including those that use precipitation or aridity rather than soil moisture (Sun et al., 2026) and single experiment studies with fewer species (e.g., Rollinson and Kaye, 2012; Castillioni et al., 2022). A growing body of evidence suggests that instead, species traits may be more predictive of plant responses to environmental change (e.g., Díaz et al., 2016). Major traits related to root

and leaf structure can impact species' drought tolerance. For trees, traits related to drought tolerance may co-vary with frost-risk, as ring-porous species are generally more drought-tolerance but risk greater damage from spring frosts compared to diffuse porous species (Bader et al., 2022; Wang et al., 2022*a*); not surprisingly ring-porous species leaf out later then diffuse-porous species Lechowicz (1984). Such interconnections between phenology and other traits have been described in other systems, too (e.g., Ocheltree et al., 2020), and suggest the potential for a framework to better predict the high variability of phenology responses across species (e.g., Morales-Castilla et al., 2024).

Our finding that sensitivity to soil moisture and interactions with temperature varies across phenophases aligns with recent studies, as well. Buonaiuto and Wolkovich (2021), for example, also found varying sensitivity of flower and leaf phenology to environmental cues, even within species. In our study, variability across species in the soil moisture response was lowest for budburst (2.7 compared to 3.8 for leafout and 3.8 for flowering), perhaps suggesting that, across species, soil moisture is a key control on timing of budburst (e.g., by affecting cell elongation (Essiamah and Eschrich, 1986)). The direction of interactive effects of soil moisture and temperature also varied in sign across phenophases, with weak positive estimates for budburst and leafout, and stronger negative interactive effects for flowering (Fig. 1). Thus, the implications of climate change driven shifts in soil moisture for phenology are likely to depend on when during the growing season shifts are greatest and, especially for flowering, how it intersects with temperature shifts, among other factors.

Forecasting multiple drivers

Soil moisture is and will continue to shift with climate change (Berg et al., 2017). Thus, while we found soil moisture had a smaller effect size than temperature (Fig. 1), it could have important implications for phenology. These effects will depend strongly on the forecasted climate change, for precipitation and the many factors that in turn impact soil moisture. How dry soils will become depends on precipitation amounts, but also rising temperatures, changes in humidity and how much water different soils can hold (CITES). Combining these factors into broad regional scales, current forecasts suggest that some areas, such as ... Some areas, such as the northeastern United States (where many of the experiments were conducted) are getting wetter, and other places are expected to get drier (Berg et al., 2017). Overall, our forecasting suggests that temperature will continue to be a dominant controller of phenology, but that soil moisture also matters, especially for certain species.

The experimental data we synthesize here highlight that multiple drivers that are shifting with global change affect phenology. Highly-cited phenology research in temperate grassland and forest systems has frequently ignored effects of soil moisture and other drivers, focusing instead on temperature. But warming can interact with other global change drivers; indeed, both synergistic and antagonistic interactions between warming and other global changes, including precipitation, were found to influence plant phenology in a recent meta-analysis of experimental studies (Zhou et al., 2023),

Forecasting of phenology shifts with global change may be made more accurate, then, by incorporating the multiple drivers of phenology that have been identified in experimental studies. To do this forecasting, we need to improve how we relate experiments to the 'real world'. This includes moving beyond treatments levels to analyze plot-level microclimate- closer to how plants may be experiencing treatments. Our study differs from some because we used field-measured soil moisture, whereas most studies use precipitation (e.g., Tao et al., 2020) or gridded moisture products (e.g., Tao et al., 2019). The problems with these proxies are widely known, including challenges with validation at fine spatiotemporal resolutions, though a number of new products are available and seem promising (Peng et al., 2021; Brocca et al., 2024). Soil moisture has not been a focus of previous phenology meta-analyses (e.g., Wolkovich et al., 2013; **?**; Zhou et al., 2023),

nor of most multi-species phenology studies in temperate mesic grasslands and forest ecosystems (e.g., Vitasse et al., 2021), despite the reality that many experiments collect these data.

Our findings offer robust, experimental evidence that builds on large-scale observational research, which increasingly suggests an important role for soil moisture in phenology (e.g., Luo et al., 2021). We focused on soil moisture because of how relevant it is to plant growth and development, especially compared to other metrics such as precip Campanella et al. (2020); Cui et al. (2022); Zhang et al. (2023). Soil moisture, however, is much harder to measure and our focus on measured soil moisture did create a data limitation as we were able to use only a subset of all the climate change experiments included in the ExPhen and MC3E databases. The fact that we could use only six of the 37 studies in STONE (16%) highlights the need for increased measurement, reporting, and sharing of environmental conditions—including soil moisture and temperature– alongside phenology observations. Such increased data could help fully disentangle how temperature is affected by soil moisture, and how soil moisture is affected by experimental temperature treatments, to drive better forecasts.

The soil moisture-phenology relationships we quantify across species may scale up to result in community-level shifts in phenology patterns. And further, may interact with other factors to affect ecosystem-level functions such as biomass and carbon uptake or storage. Disentangling effects from climate versus soil can be challenging, but has demonstrated the importance of species interactions and multiple limiting nutrients in vegetation responses to global change (Wilfahrt et al., 2021). In addition to playing a role in budburst, leafout, and flowering phenology, for example, soil moisture affects plant resorption of nitrogen and phosphorus during leaf senescence(Estiarte et al., 2022). Shifts in soil moisture and cascading effects on phenology may ultimately lead to changes in growing season length and carbon uptake (Wolkovich et al., 2025), especially since soil moisture is a key control of nutrient cycling, including nitrogen and carbon (Liu et al., 2019).

Conclusions

Now underway for four decades, climate change experiments (e.g., Tamaki et al., 1981; Carlson and Bazzaz, 1982; Melillo et al., 2017) can provide a unique opportunity to disentangle multiple drivers, understand biological responses to climate change, and inform global climate forecasting models. Yet the full range of changes in environmental conditions imposed by these experiments is rarely presented. Using two databases that compile microclimate data and phenology responses from multiple warming experiments we show that soil moisture, in addition to temperature, affects plant phenology. We quantified phenology delays with soil drying across budburst, leafout, and flowering that suggest these effects should be more often included in modelling and forecasts of seasonal shifts with climate change. The specific impacts of including moisture when forecasting phenology, however, are likely to be context dependent, especially compared to the effects of temperature, where climate change is almost always expected to increase temperatures and thus advance spring phenology in most temperate systems. In contrast, our results combined with the complexity of forecasting precipitation and drying, suggest shifts in soil moisture will result in more muted and also more exaggerated phenology shifts, depending on the regional climate forecast and the species present. Managers will thus need locally relevant biological and climate information to apply our findings to assess impacts of climate change and manage natural resources for enhanced climate resilience.

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Figures

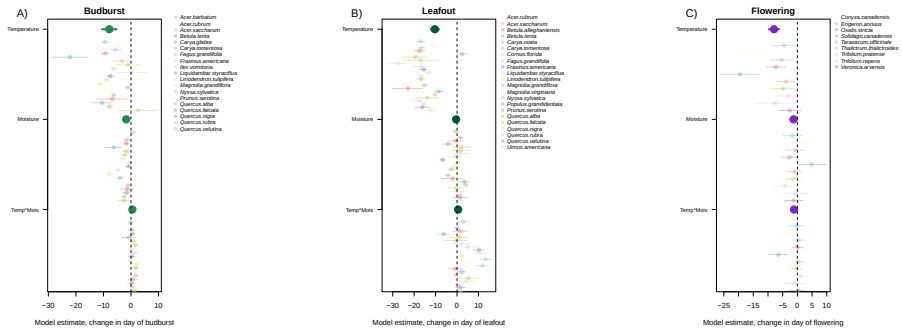


Figure 1: **Model coefficients from budburst, leafout, and flowering models (with standardized predictors)** for the most common species (i.e., those at more than one study site). Plot of species effects across all species available in Supplemental Materials, Fig. S3.

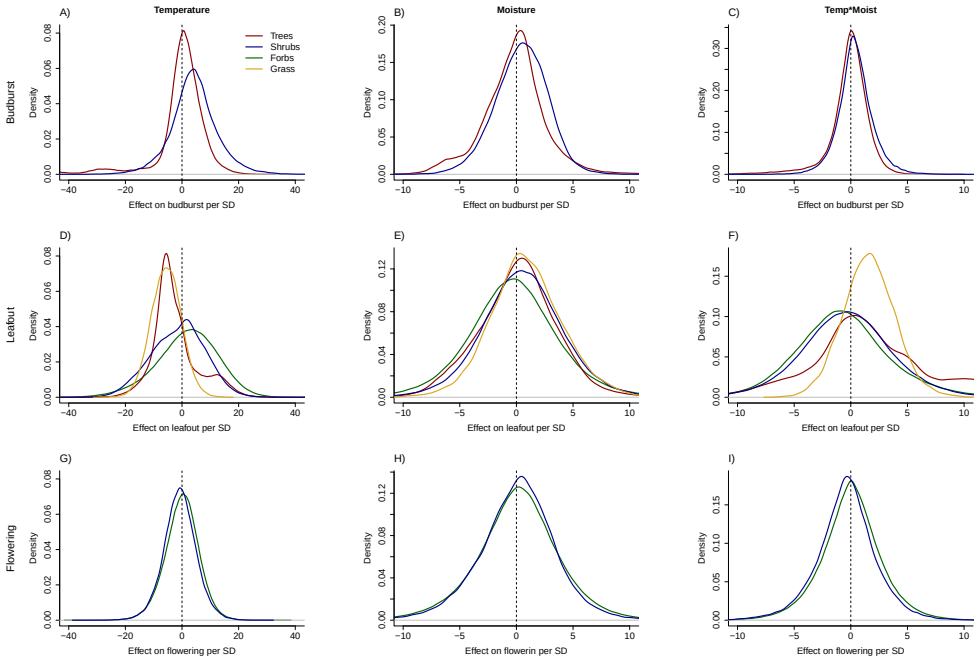


Figure 2: **Effects of temperature, soil moisture, and their interaction, summarized by life forms** on budburst (A-C), leafout (D-F), and flowering (G-I) reveal minimal differences in estimated responses to soil moisture (middle column, panels B,E,H). Curves show probability density functions for posterior samples of estimated effects for temperature, soil, and their interaction summarized by species and grouped into four life forms (trees, shrubs, forbs, and grasses). Patterns do appear to vary by life form for temperature (first column) and interactions (third column). For budburst, temperature effects were more negative for trees compared to shrubs, and more negative for both trees and grasses compared to shrubs and forbs for leafout. Interactions between temperature and moisture effects on leafout, on the other hand, skew more positive for grasses compared to other life-forms.

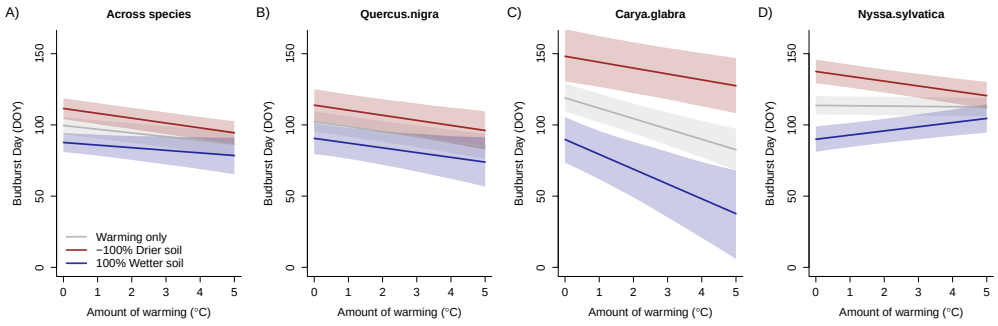


Figure 3: **Patterns of forecasted changes in budburst date with warming vary with soil moisture and across species**. Across all species, our model estimated negative effects (i.e., earlier) of both temperature and soil moisture on budburst and a weak interaction between the two effects (A, and example species *Quercus nigra* in B); however, the magnitude of these effects, as well as the sign and magnitude of the estimated interaction between soil moisture and temperature, differed across species, resulting in divergent patterns with warming under different soil moisture regimes. Budburst may occur much earlier in wetter vs drier soils with warming for species that have a synergistic interaction between soil moisture and temperature, such as *Carya glabra* (C). Whereas, other species with an antagonistic interaction, such as *Nyssa sylvatica*(D), may experience delayed budburst in wet soils but advance in dry soils.